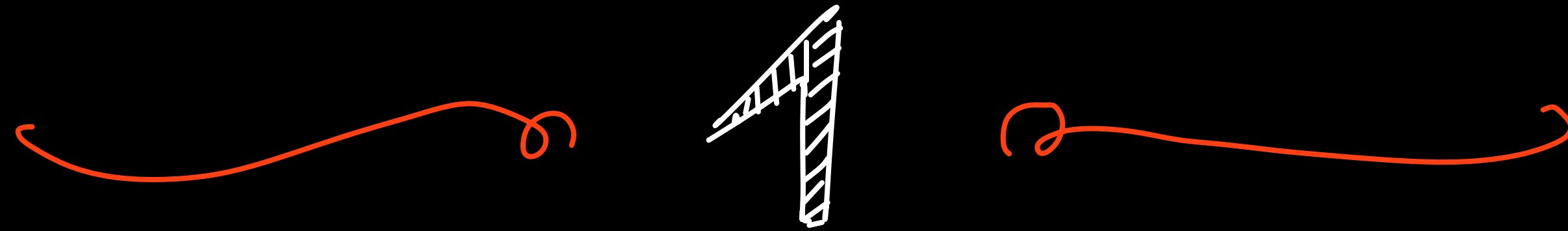


Interpolation method: **old** and **new**

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Existence of the thermodynamic limit of the free energy for the Sherrington-Kirkpatrick spin glass

(Guerra - Toninelli 02)

"Interpolation method"

SK model: $H_N^J(\sigma) = \frac{1}{\sqrt{N}} \sum_{1 \leq i < j \leq N} J_{ij} \sigma_i \sigma_j \quad \left(+ h \sum_{i=1}^N \sigma_i \right)$
 (S-K 75')
 $\sigma \in C_N \doteq \{-1, 1\}^N \quad J_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$

Gibbs measure: $P_N^J(\sigma) = \frac{e^{H_N^J(\sigma)}}{Z_N^J}, \quad Z_N^J = \sum_{\sigma \in C_N} e^{H_N^J(\sigma)}$

$$\langle A(\sigma^1, \dots, \sigma^k) \rangle = \sum_{\sigma^1 \in C_N} \dots \sum_{\sigma^k \in C_N} A(\sigma^1, \dots, \sigma^k) \prod_{j=1}^k P_N^J(\sigma^j)$$

$\sigma^1, \dots, \sigma^k$ conditionally (on J) iid replicas.

Free energy: $f_N = \frac{1}{N} \mathbb{E}_J \ln \sum_{\sigma \in C_N} e^{H_N^J(\sigma)}$

Thm (Guerra - Toninelli 02):

The thermodynamic limit of the free energy exists:

$$\lim_{N \rightarrow +\infty} f_N = f$$

proof:

$$H_N^t(\sigma) = \sqrt{\frac{t}{N}} \sum_{1 \leq i < j \leq N} J_{ij} \sigma_i \sigma_j$$

$$+ \sqrt{\frac{1-t}{N_1}} \sum_{1 \leq i < j \leq N_1} J'_{ij} \sigma_i \sigma'_j + \sqrt{\frac{1-t}{N_2}} \sum_{N_1 < i < j \leq N_2} J''_{ij} \sigma_i \sigma_j$$

$$N = N_1 + N_2$$

$$J_{ij}, J'_{ij}, J''_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

$$P_N^t(\sigma) \propto e^{H_N^t(\sigma)}$$

$$\underbrace{\left[\frac{N_1}{SK} \right]}_{SK} \underbrace{\left[\frac{N_2}{SK} \right]}_{SK} 0 \xrightarrow{t} 1 \underbrace{\left[\frac{N}{SK} \right]}_{SK}$$

interpolating free energy: $f_N^t = \frac{1}{N} \mathbb{E}_{J, J', J''} \ln \sum_{\sigma \in \mathcal{C}_N} e^{H_N^t(\sigma)}$

boundary values:

$$f_N^{t=1} = A_N \quad \boxed{N}$$

$$f_N^{t=0} = \frac{1}{N} E \ln \left(\sum_{\sigma_1 \in C_{N_1}} e^{H_{N_1}^{\sigma_1}(\sigma_1)} \sum_{\sigma_2 \in C_{N_2}} e^{H_{N_2}^{\sigma_2}(\sigma_2)} \right)$$

$$= \frac{N_1}{N} \frac{1}{N_1} E \ln \sum_{\sigma_1 \in C_{N_1}} e^{H_{N_1}^{\sigma_1}(\sigma_1)} + \frac{N_2}{N} \frac{1}{N_2} E \ln \sum_{\sigma_2 \in C_{N_2}} e^{H_{N_2}^{\sigma_2}(\sigma_2)}$$

$$= \frac{N_1}{N} f_{N_1} + \frac{N_2}{N} f_{N_2}$$

$$\boxed{N_1} \quad \boxed{N_2}$$

If: $f_N \geq \frac{N_1}{N} f_{N_1} + \frac{N_2}{N} f_{N_2}$

$\Rightarrow f_N$ is superadditive $\Rightarrow \lim_{N \rightarrow +\infty} f_N$ exists

(by Fekete's lemma)

Thus: $f_N^{t=1} \geq f_N^{t=0}$

YES if

$$\boxed{\frac{df_N^t}{dt} \geq 0}$$

$$\frac{df_N^t}{dt} \approx \frac{d}{dt} \frac{1}{N} \mathbb{E} \ln \sum_{\sigma \in \mathcal{C}_N} e^{H_N^t(\sigma)}$$

$$P_N^t(\sigma)$$

$$= \frac{1}{N} \mathbb{E} \frac{\sum_{\sigma} \frac{dH_N^t(\sigma)}{dt} e^{H_N^t(\sigma)}}{\sum_{\sigma} e^{H_N^t(\sigma)}} = \frac{1}{N} \mathbb{E} \left\langle \frac{dH_N^t(\sigma)}{dt} \right\rangle_t$$

$$= \frac{1}{2N} \mathbb{E} \left[\underbrace{\frac{1}{\sqrt{t}N} \sum_{1 \leq i < j \leq N} J_{ij} \langle \sigma_i \sigma_j \rangle_t}_{?} - \frac{1}{\sqrt{(1-t)N_1}} \sum_{1 \leq i < j \leq N_1} J_{ij}^I \langle \sigma_i \sigma_j \rangle_t - \frac{1}{\sqrt{(1-t)N_2}} \sum_{N_1 < i < j \leq N} J_{ij}^{II} \langle \sigma_i \sigma_j \rangle_t \right]$$

Gaussian IPP formula: $\mathbb{E}(J g(J)) = \mathbb{E}(g'(J))$

$$J \sim \mathcal{CP}(0, 1)$$

$$\frac{1}{2N} \sum_{1 \leq i < j \leq N} \mathbb{E} \left[\frac{1}{\sqrt{Nt}} J_{ij} \langle \phi_i \phi_j \rangle_t \right] \quad e \left(\sqrt{\frac{t}{N}} \sum_{1 \leq i < j \leq N} J_{ij} \phi_i \phi_j + \dots \right)$$

$$= \frac{1}{2N} \sum_{i < j} \frac{1}{N} \mathbb{E} \left[\langle (\phi_i \phi_j)^2 \rangle_t - \langle \phi_i \phi_j \rangle_t^2 \right]$$

$$= \frac{1}{4N^2} \sum_{i, j=1}^N \mathbb{E} \left[1 - \underbrace{\langle \phi_i \phi_j \rangle_t^2}_{\langle \phi_i^2 \phi_j^2 \rangle_t} \right]$$

$$= \frac{1}{4} \mathbb{E} \left\langle 1 - \left(\frac{1}{N} \sum_{i=1}^N \phi_i^2 \right)^2 \right\rangle_t$$

overlaps:

$$Q = \frac{1}{N} \sum_{i=1}^N \phi_i^1 \phi_i^2$$

$$Q^{(1)} = \frac{1}{N_1} \sum_{i=1}^{N_1} \phi_i^1 \phi_i^2$$

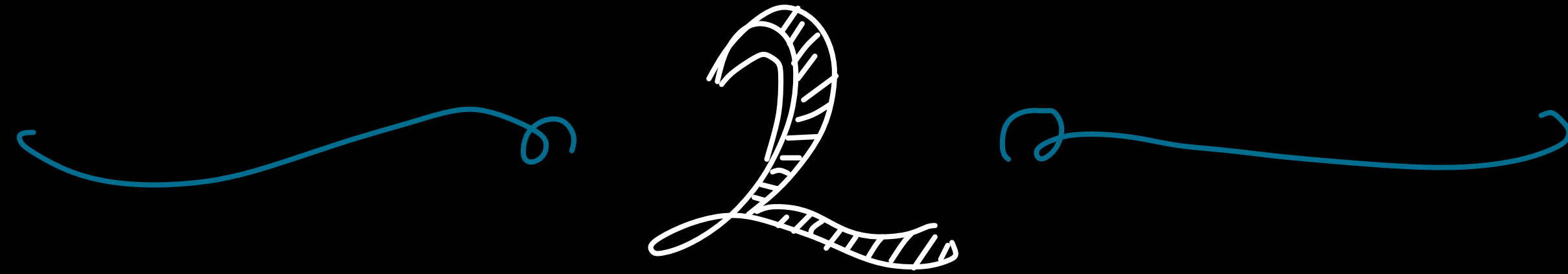
$$Q^{(2)} = \frac{1}{N_2} \sum_{i=N_1+1}^N \phi_i^1 \phi_i^2$$

$$\Rightarrow \frac{dE_N^t}{dt} = -\frac{1}{4} E \left\langle Q^2 - \frac{N_1}{N} (Q^{(1)})^2 - \frac{N_2}{N} (Q^{(2)})^2 \right\rangle_t$$

$x \mapsto x^2$ convex and $Q = \frac{N_1}{N} Q^{(1)} + \frac{N_2}{N} Q^{(2)}$

$$\Rightarrow Q^2 - \frac{N_1}{N} (Q^{(1)})^2 - \frac{N_2}{N} (Q^{(2)})^2 \leq 0$$

$$\Rightarrow \frac{dE_N^t}{dt} \geq 0 \quad \text{QED}$$



Single-letter formula for the mutual information of the spike Wigner model (or symmetric rank-one matrix factorization)

(Barbier - Macris 16')

"Adaptive interpolation method"

Spike Wigner model:
(Johnstone 01')

$$Y_{ij} = \sqrt{\frac{\lambda}{N}} \tilde{x}_i^* \tilde{x}_j^* + Z_{ij} \quad 1 \leq i < j \leq N$$
$$\tilde{x}_i^* \stackrel{\text{iid}}{\sim} P_0, \quad Z_{ij} = Z_{ji} \stackrel{\text{iid}}{\sim} \mathcal{N}(0,1)$$

Thm (Korada-Macris 09', Deshpande-Abbe-Montanari 16'
Barbier-Dia-Macris-Krzakala-Zdeborová 16'
Miolane-Lelarge 16', Barbier-Macris 16'):

$$\lim_{N \rightarrow \infty} \frac{1}{N} I(Y; X^*) = \inf_{q \geq 0} \left\{ \frac{\lambda}{4} \left[q - \underbrace{\mathbb{E}((x_1^*)^2)}_{=1} \right]^2 + I(x_1^*; \sqrt{\lambda q} x_1^* + Z) \right\}$$
$$x_1^* \sim P_0 \quad Z \sim \mathcal{N}(0,1)$$

proof:

interpolating inference model:

$$\begin{cases} y_{ij}(t) = \sqrt{(1-t) \frac{\lambda}{N}} x_i^* x_j^* + z_{ij} \\ \tilde{y}_i(t, \epsilon) = \sqrt{R(t, \epsilon)} x_i^* + \tilde{z}_i \end{cases}$$

$$R(t, \epsilon) = \epsilon + \lambda \int_0^t q(s, \epsilon) ds, \quad R(0, \epsilon) = \epsilon$$

$$z_{ij} = z_{ji}, \quad \tilde{z}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

$$l_N(t, \epsilon) \stackrel{\circ}{=} \frac{1}{N} \mathbb{I}((y(t), \tilde{y}(t, \epsilon)); x^*) \quad \boxed{y = \sqrt{\frac{\lambda}{N}} X X^T + z} \quad 0 \xrightarrow{t} 1 \quad \boxed{\tilde{y} = \sqrt{R(1, \epsilon)} X + z}$$

SW model
(+ ϵ -correction) AWGN "decoupled" model

posterior: $dP(x | y(t), \tilde{y}(t, \epsilon)) = \frac{1}{\mathcal{Z}(y(t), \tilde{y}(t))} \prod_{i \leq N} dP_0(x_i)$

$$\times \exp \left\{ -\frac{1}{2} \sum_{i,j} \left(y_{ij}(t) - \sqrt{(1-t) \frac{\lambda}{N}} x_i x_j \right)^2 - \frac{1}{2} \left\| \tilde{y}(t, \epsilon) - \sqrt{R(t, \epsilon)} X \right\|^2 \right\}$$

posterior expectation: $\langle A(x) \rangle_t \stackrel{\circ}{=} \int dP(x | y(t), \tilde{y}(t, \epsilon)) A(x)$

proof:

interpolating inference model:

$$\begin{cases} Y_{ij}(t) = \sqrt{(1-t) \frac{\lambda}{N}} X_i^* X_j^* + z_{ij} \\ \tilde{Y}_i(t, \epsilon) = \sqrt{R(t, \epsilon)} X_i^* + \tilde{z}_i \end{cases}$$

$$R(t, \epsilon) = \epsilon + \lambda \int_0^t q(s, \epsilon) ds, \quad R(0, \epsilon) = \epsilon$$

$$z_{ij} = z_{ji}, \quad \tilde{z}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

$$l_N(t, \epsilon) \stackrel{\circ}{=} \frac{1}{N} \mathbb{I}((Y(t), \tilde{Y}(t, \epsilon)); X^*) \quad \boxed{Y = \sqrt{\frac{\lambda}{N}} X X^T + z} \quad 0 \xrightarrow{t} 1 \quad \boxed{\tilde{Y} = \sqrt{R(1, \epsilon)} X + z}$$

SW model (+ ϵ -correction) AWGN "decoupled" model

Boundary values:

$$l_N(t=0, \epsilon) = \frac{1}{N} \mathbb{I}(Y; X^*) + o(\epsilon)$$

$\frac{1}{N} \mathbb{I}(\sqrt{\epsilon} X^* + z; X^*)$

$$l_N(t=1, \epsilon) = \mathbb{I}\left(\sqrt{\lambda \int_0^1 q(t, \epsilon) dt} X_1^* + \tilde{z}_1; X_1^*\right) + o(\epsilon)$$

$\mathbb{I}(\sqrt{\gamma} X^* + z; X^*) \quad \sigma/2\text{-Lipschitz}$

Sum rule: $\frac{1}{N} I(\gamma; x^*) = i^{\text{Pot}} \left(\int_0^1 q(t, \epsilon) dt \right) + \frac{\lambda}{4} (\mathcal{R}_1 - \mathcal{R}_2 - \mathcal{R}_3) + o(\epsilon) + o(N^{-1})$

$i^{\text{Pot}}(q) \stackrel{0}{=} \frac{\lambda}{4} (q - \epsilon)^2 + I(x_i^*; \sqrt{\lambda} q x_i^* + z)$ RS potential

$$\left\{ \begin{array}{ll} \mathcal{R}_1 \stackrel{0}{=} \int_0^1 \left(q(t, \epsilon) - \int_0^1 q(s, \epsilon) ds \right)^2 dt & \text{"path fluctuation"} \\ \mathcal{R}_2 \stackrel{0}{=} \int_0^1 \mathbb{E} \gamma(t), \tilde{\gamma}(t, \epsilon) \left\langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \right\rangle_t & \text{"overlap fluct."} \\ \mathcal{R}_3 \stackrel{0}{=} \int_0^1 \left(q(t, \epsilon) - \mathbb{E} \langle Q \rangle_t \right)^2 dt & \text{"path mismatch"} \end{array} \right.$$

overlap: $Q \stackrel{0}{=} \frac{1}{N} \sum_{i=1}^N x_i x_i^*$

$$\dot{i}_N(0, \epsilon) = i_N(1, \epsilon) - \int_0^1 \frac{d i_N(t, \epsilon)}{dt} dt$$

$$\frac{1}{N} I(\gamma; x^*) = I(\sqrt{\lambda \int_0^1 q(t, \epsilon) dt} x_1^* + \tilde{z}_1; x_1^*) + o(\epsilon) - \int_0^1 i_N^1 dt$$

$$\frac{d}{dt} \frac{1}{N} I(\gamma(t), \tilde{\gamma}(t, \epsilon); x^*) = 2 \quad \begin{cases} \gamma_{ij}(t) = \sqrt{(1-t) \frac{\lambda}{N}} x_i^* x_j^* + z_{ij} \\ \tilde{\gamma}_i(t, \epsilon) = \sqrt{R(t, \epsilon)} x_i^* + \tilde{z}_i \end{cases}$$

I-MMSE: $\gamma = \sqrt{\lambda} g(x^*) + z \quad z \sim \mathcal{N}(0, I_d)$

then: $\frac{d}{d\lambda} I(\gamma; x^*) = \frac{1}{2} \mathbb{E}_{x^*, \gamma} \|g(x^*) - \langle g(x) \rangle\|^2$

$$\frac{d i_N(t, \epsilon)}{dt} = - \frac{\lambda}{4} \frac{1}{N^2} \mathbb{E} \|x^* x^{*T} - \langle x x^T \rangle_t\|_F^2 + \frac{\lambda}{2} q(t, \epsilon) \frac{1}{N} \mathbb{E} \|x^* - \langle x \rangle_t\|^2 + o(N^{-1})$$

$\frac{d}{dt} R(t, \epsilon)$

Nishimori id. $\mathbb{E}_{x^*} \mathbb{E}_{y|x^*} \mathbb{E}_{x^1|y} g(x^*, x^1) \stackrel{\text{Bayes}}{=} \mathbb{E}_y \mathbb{E}_{x^*|y} \mathbb{E}_{x^1|y} g(x^*, x^1)$

$$\mathbb{E} \langle g(x^*, x^1) \rangle = \mathbb{E} \langle g(x^2, x^1) \rangle$$

ex: $\frac{1}{N} \mathbb{E} \|x^* - \underbrace{\langle x \rangle_t}_{\mathbb{E}[x|y(t), \tilde{y}(t)]}\|^2 = \frac{1}{N} \left(\mathbb{E} \|x^*\|^2 + \underbrace{\mathbb{E} \|\langle x \rangle_t\|^2}_{= \mathbb{E}[\langle x \rangle_t \cdot \langle x \rangle_t]} - 2 \mathbb{E}[x^* \cdot \langle x \rangle_t] \right)$

$$= \mathbb{E} \langle x^1 \cdot x^2 \rangle_t = \mathbb{E} [x^* \cdot \langle x \rangle_t]$$

$$= \underbrace{\mathbb{E} [(x_i^*)^2]}_p - \mathbb{E} \langle Q \rangle_t \quad Q = \frac{1}{N} \sum_{i=1}^N x_i^* x_i$$

Similarly: $\frac{1}{N^2} \mathbb{E} \|x^* x^{*T} - \langle x x^T \rangle_t\|^2 = p^2 - \mathbb{E} \langle Q^2 \rangle_t$

Sum rule: $\frac{1}{N} \mathbb{I}(\gamma; x^*) = i^{\text{pot}}(\int_0^1 q(t, \epsilon) dt) + \frac{\lambda}{4} (\mathcal{R}_1 - \mathcal{R}_2 - \mathcal{R}_3) + o(\epsilon) + o(\frac{1}{N})$

$$\begin{cases} \mathcal{R}_1 \doteq \int_0^1 (q(t, \epsilon) - \int_0^1 q(s, \epsilon) ds)^2 dt \\ \mathcal{R}_2 \doteq \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t \\ \mathcal{R}_3 \doteq \int_0^1 (q(t, \epsilon) - \mathbb{E} \langle Q \rangle_t)^2 dt \end{cases}$$

Upper bound: Fix $q(t, \epsilon) = q \geq 0$ and $\epsilon = 0 \Rightarrow \mathcal{R}_1 = 0$
 "Linear path"

Moreover: $\mathcal{R}_2, \mathcal{R}_3 \geq 0$

$$\Rightarrow \limsup_{N \rightarrow \infty} \frac{1}{N} \mathbb{I}(\gamma; x^*) \leq \inf_{q \geq 0} i^{\text{pot}}(q)$$

Sum rule: $\frac{1}{N} \mathbb{I}(Y; X^*) = i^{\text{pot}} \left(\int_0^1 q(t, \epsilon) dt \right) + \frac{\lambda}{4} (\mathcal{R}_1 - \mathcal{R}_2 - \mathcal{R}_3) + o(\epsilon) + o\left(\frac{1}{N}\right)$

$$\begin{cases} \mathcal{R}_1 \doteq \int_0^1 \left(q(t, \epsilon) - \int_0^1 q(s, \epsilon) ds \right)^2 dt \\ \mathcal{R}_2 \doteq \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t \\ \mathcal{R}_3 \doteq \int_0^1 \left(q(t, \epsilon) - \mathbb{E} \langle Q \rangle_t \right)^2 dt \end{cases}$$

Lower bound: $\mathcal{R}_1 \geq 0 \Rightarrow \mathcal{R}_2, \mathcal{R}_3 \sim 0^2$

"adaptive path"

$$\mathcal{R}_3 = 0 \text{ implies } q(t, \epsilon) = \mathbb{E} \langle Q \rangle_{t, \epsilon} \quad (\Rightarrow \quad \left(\begin{array}{l} \frac{dR(t, \epsilon)}{dt} = f(t, R(t, \epsilon)) \\ \text{with } R(0, \epsilon) = \epsilon \end{array} \right) \quad \text{ODE})$$

$$\mathbb{E} \langle Q \rangle_{t, \epsilon} \leq \mathbb{E}[(X_1^*)^2] \text{ bounded}$$

$\Rightarrow$ Cauchy-Lipschitz thm $\Rightarrow \exists$ unique global solution $R^*(t, \epsilon)$

Sum rule: $\frac{1}{N} \mathbb{I}(\gamma; x^*) = i^{\text{pot}} \left(\int_0^1 q(t, \epsilon) dt \right) + \frac{\lambda}{4} (\mathcal{R}_1 - \mathcal{R}_2 - \mathcal{R}_3) + o(\epsilon) + o\left(\frac{1}{N}\right)$

$$\begin{cases} \mathcal{R}_1 \doteq \int_0^1 (q(t, \epsilon) - \int_0^1 q(s, \epsilon) ds)^2 dt \\ \mathcal{R}_2 \doteq \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t \\ \mathcal{R}_3 \doteq \int_0^1 (q(t, \epsilon) - \mathbb{E} \langle Q \rangle_t)^2 dt \end{cases}$$

Lower bound:
"adaptive path"

$R = R^*(t, \epsilon) \Rightarrow \mathcal{R}_3 = 0.$ What about $\mathcal{R}_2$?

Proposition: If $\frac{dR^*(t, \epsilon)}{d\epsilon} \geq 1$ then $\frac{1}{S_N} \int_{S_N}^{2S_N} \mathcal{R}_2 d\epsilon \xrightarrow{N \rightarrow \infty} 0$

"Replica symmetry"

"regularity condition"

$$R_2 = \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t$$

"thermal" fluctuations: $F(R) \doteq \frac{1}{N} \ln \mathcal{Z}(\gamma(t), \tilde{\gamma}(t))$ "free energy"

$$\left\{ \begin{aligned} \frac{dF}{dR} &= \left\langle \frac{d \ln \mathcal{Z}}{dR} \right\rangle_t = \left\langle \frac{d}{dR} \mathcal{Z}(\times 1 \gamma(t), \tilde{\gamma}(t)) \right\rangle = \langle \mathcal{Z}' \rangle = \underline{\underline{O(1)}} \\ &\quad \uparrow \text{log-likelihood} \\ \frac{1}{N} \frac{d^2 F}{dR^2} &= - \langle (\mathcal{Z}' - \langle \mathcal{Z}' \rangle_t)^2 \rangle_t + O(1/N) \end{aligned} \right.$$

$$\int_{S_N}^{2S_N} dt \langle (\mathcal{Z}' - \langle \mathcal{Z}' \rangle_t)^2 \rangle_t \stackrel{!}{=} \int_{R(t, S_N)}^{R(t, 2S_N)} dR \langle - \rangle_t + O(1/N)$$

$$\frac{dR}{dt} \geq 1 \quad \leq \frac{1}{N} \left(\frac{dF}{dR} \right)_{R(t, S_N)}^{R(t, 2S_N)} + O(1/N) = O(1/N)$$

$$R_2 = \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t$$

"quenched" fluctuations: $\int dt \mathbb{E} [(\langle Z' \rangle_t - \mathbb{E} \langle Z' \rangle_t)^2] = o_N(1)$

$$\frac{dF}{dR} = \langle Z' \rangle \quad \text{and} \quad \frac{d \mathbb{E} F}{dR} = \mathbb{E} \langle Z' \rangle$$

1) lemma (free energy concentration): $\mathbb{E} [(F - \mathbb{E} F)^2] \xrightarrow{N \rightarrow \infty} 0$

2) convexity lemma: $F(R), \mathbb{E} F(R) \sim \text{CONCAVE} + F \rightarrow \mathbb{E} F$
 $\Rightarrow \frac{dF}{dR} \rightarrow \frac{d \mathbb{E} F}{dR}$

Finally:

by Nishimori: $\underbrace{C \mathbb{E} \langle (Z' - \mathbb{E} \langle Z' \rangle_t) \rangle_t}_{o_N(1)} \geq \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle$

Sum rule: $\frac{1}{N} \mathbb{I}(Y; X^*) = i^{\text{pot}} \left(\int_0^1 q(t, \epsilon) dt \right) + \frac{\lambda}{4} (\mathcal{R}_1 - \mathcal{R}_2 - \mathcal{R}_3) + o(\epsilon) + o\left(\frac{1}{N}\right)$

$$\begin{cases} \mathcal{R}_1 \doteq \int_0^1 \left(q(t, \epsilon) - \int_0^1 q(s, \epsilon) ds \right)^2 dt \\ \mathcal{R}_2 \doteq \int_0^1 \mathbb{E} \langle (Q - \mathbb{E} \langle Q \rangle_t)^2 \rangle_t \\ \mathcal{R}_3 \doteq \int_0^1 \left(q(t, \epsilon) - \mathbb{E} \langle Q \rangle_t \right)^2 dt \end{cases}$$

Lower bound:
"adaptive path"

Prop: If $\frac{dR^*(t, \epsilon)}{d\epsilon} \geq 1$ then $\frac{1}{S_N} \int_{S_N}^{2S_N} \mathcal{R}_2 d\epsilon \xrightarrow{N \rightarrow \infty} 0$

$$\frac{dR^*(t, \epsilon)}{d\epsilon} = \mathbb{E} \left[\int_0^1 ds \frac{d\mathbb{E} \langle Q \rangle_s}{dR} \Big| R = R^*(s, \epsilon) \right]$$

Liouville formula

$$\frac{d\mathbb{E} \langle Q \rangle_s}{dR} = \frac{1}{N} \sum_{i,j=1}^N \mathbb{E} \left[\left(\langle x_i x_j \rangle_s - \langle x_i \rangle_s \langle x_j \rangle_s \right)^2 \right] \geq 0$$

Combining everything:

$$\frac{1}{N} I(x^*; \gamma) = \frac{1}{s_N} \int_{s_N}^{2s_N} d\epsilon \quad i^{pot} \left(\int_0^1 q^*(t, \epsilon) dt \right) + O_N(1)$$

$$\geq \inf_{q \geq 0} i^{pot}(q) + O_N(1)$$

$$\Rightarrow \liminf_{N \rightarrow \infty} \frac{1}{N} I(x^*; \gamma) \geq \inf_{q \geq 0} i^{pot}(q)$$



Thank you!